

Department of Computer Science**Assignment 2****Course Code: CS-112****Name:** _____**Course Title: Object Oriented Programming****Registration No.** _____**Total Marks: 20****Discipline/Semester: BS(CS) 2nd A/B****Assigned Date: 10-04-2025****Due Date: 16-04-2025**

Instructions:

- Late submissions will not be entertained.
- Copied assignments will be marked ZERO.
- Submit the assignment in hard form
- Use of AI tool is strictly not allowed.

CLO2, GA2, C3

Question 1:

You have been assigned to develop a basic Java application for a library to help manage its book inventory. Each book in the library is described by its title, the author's name, and the number of copies currently available. In addition to storing individual book details, the system must also track the total number of book entries added to the system. To implement this:

- Create a class named Book with private instance variables for the title, author, and number of available copies.
- Use a constructor that utilizes the this keyword to initialize these fields.
- Include public getter and setter methods to access and update the values of these fields when needed.
- The class should also contain a static variable that keeps a count of all Book objects created. Every time a new book is instantiated, this count should be incremented in the constructor. Also, include a static method that returns the current total number of books in the system.
- In another class called LibraryApp(this is main class), demonstrate the functionality by creating at least three Book objects.
- Modify the number of copies for one of the books using a setter method.
- Display the details of all books using getter methods, and finally, use the static method to display the total number of books stored in the system.

Also write a brief description of the purpose of each of the main object-oriented programming constructs used in this application.